

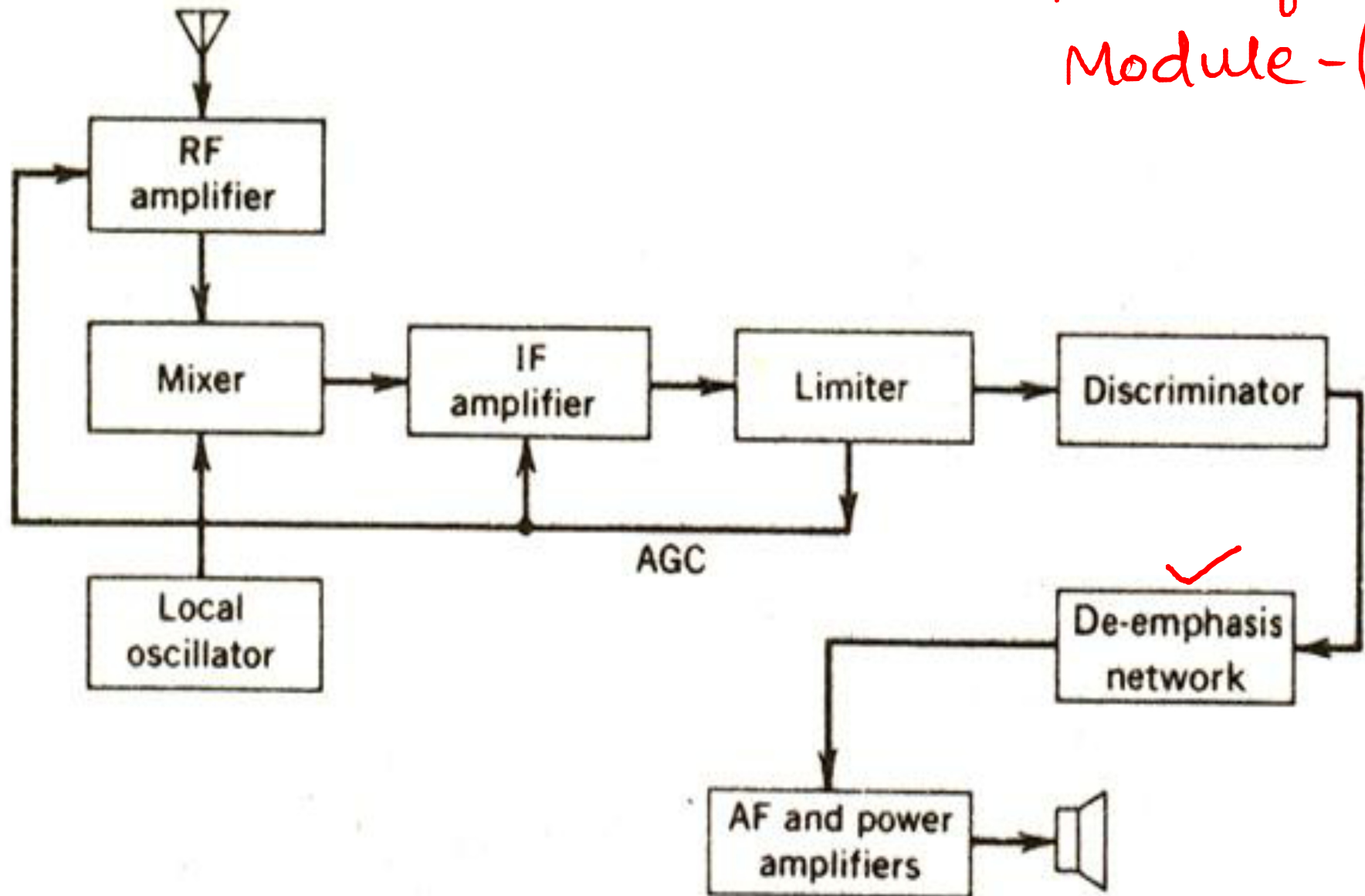


Noise Triangle, Pre-emphasis and De-emphasis

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FM Receiver

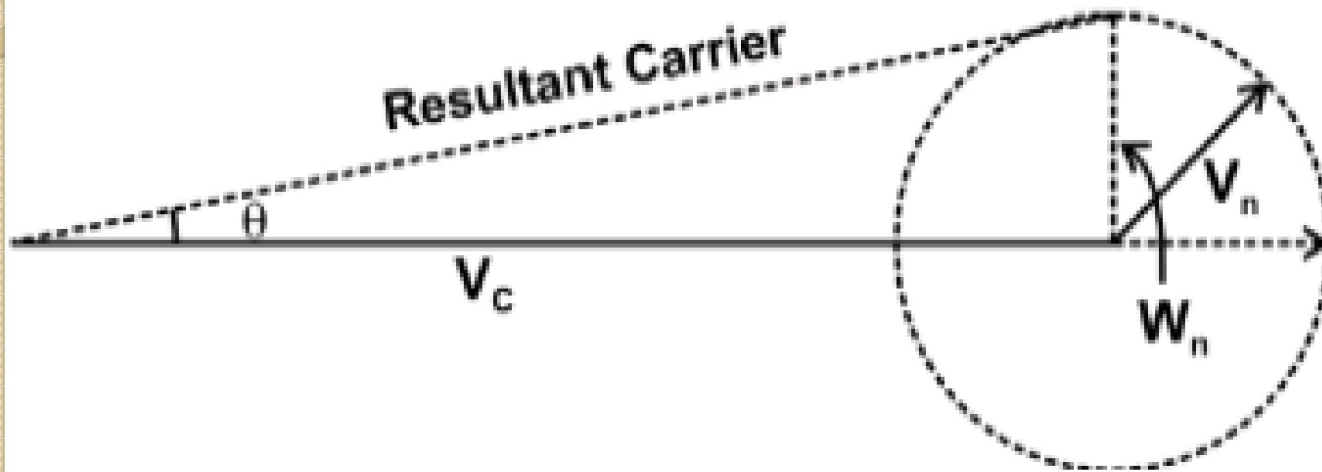
→ part of
Module - (4)



Noise Triangle

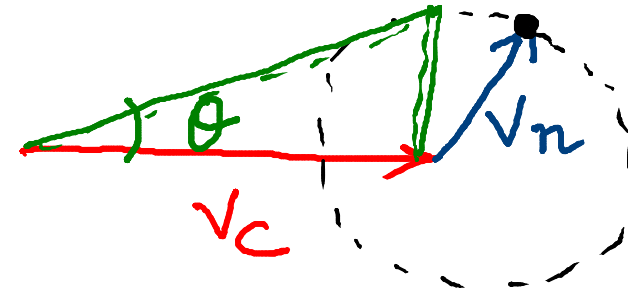
- When **thermal noise** with a constant spectral density is added to FM signal,
- It produces an **unwanted deviation** of carrier frequency
- The magnitude of this deviation depends on **relative amplitude** of the noise w.r.t. carrier

Vector effect of noise on Carrier



$$\theta = \sin^{-1} \frac{V_n}{V_c}$$

$$m = \frac{V_n}{V_c}$$

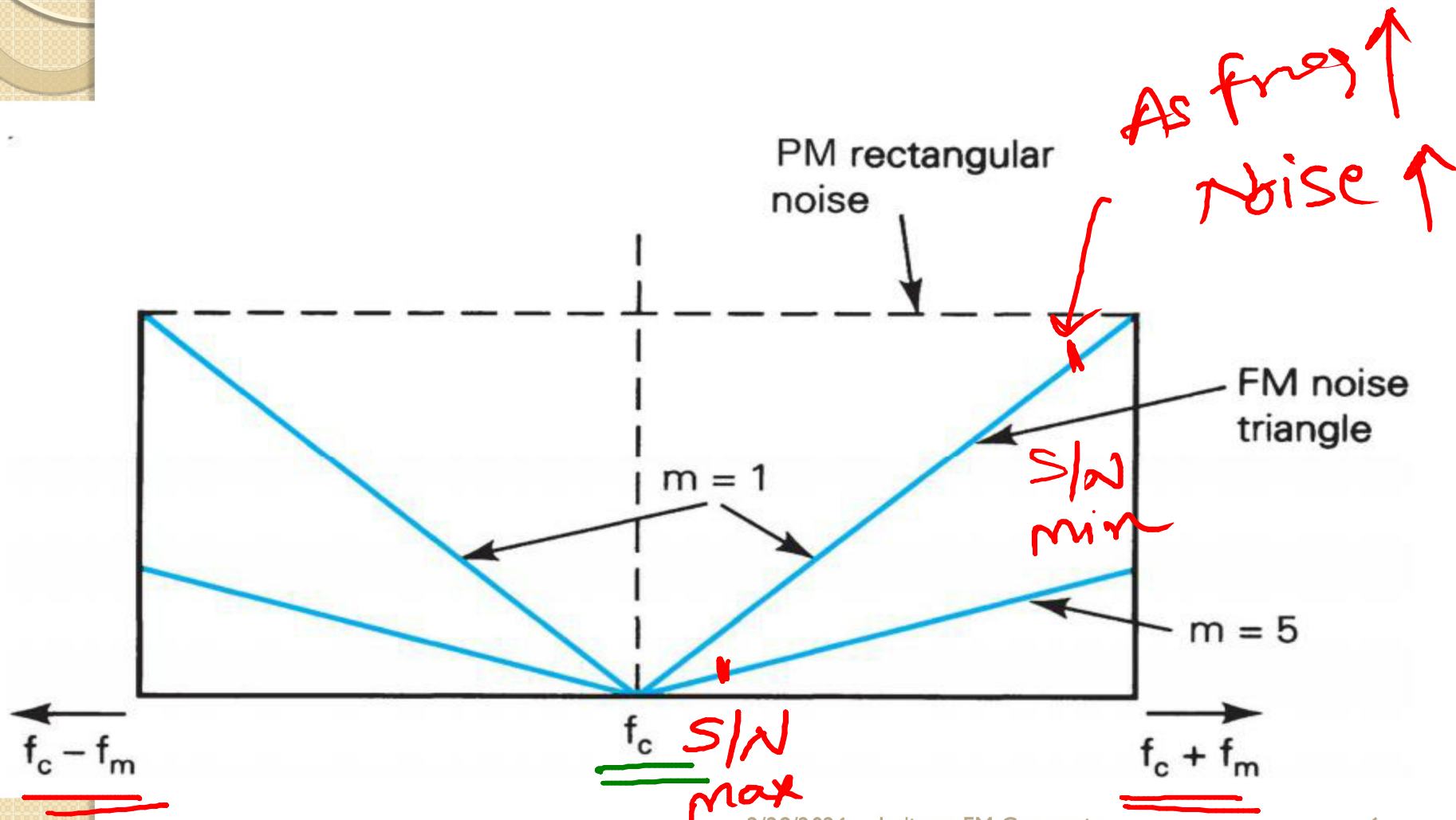


Noise Triangle

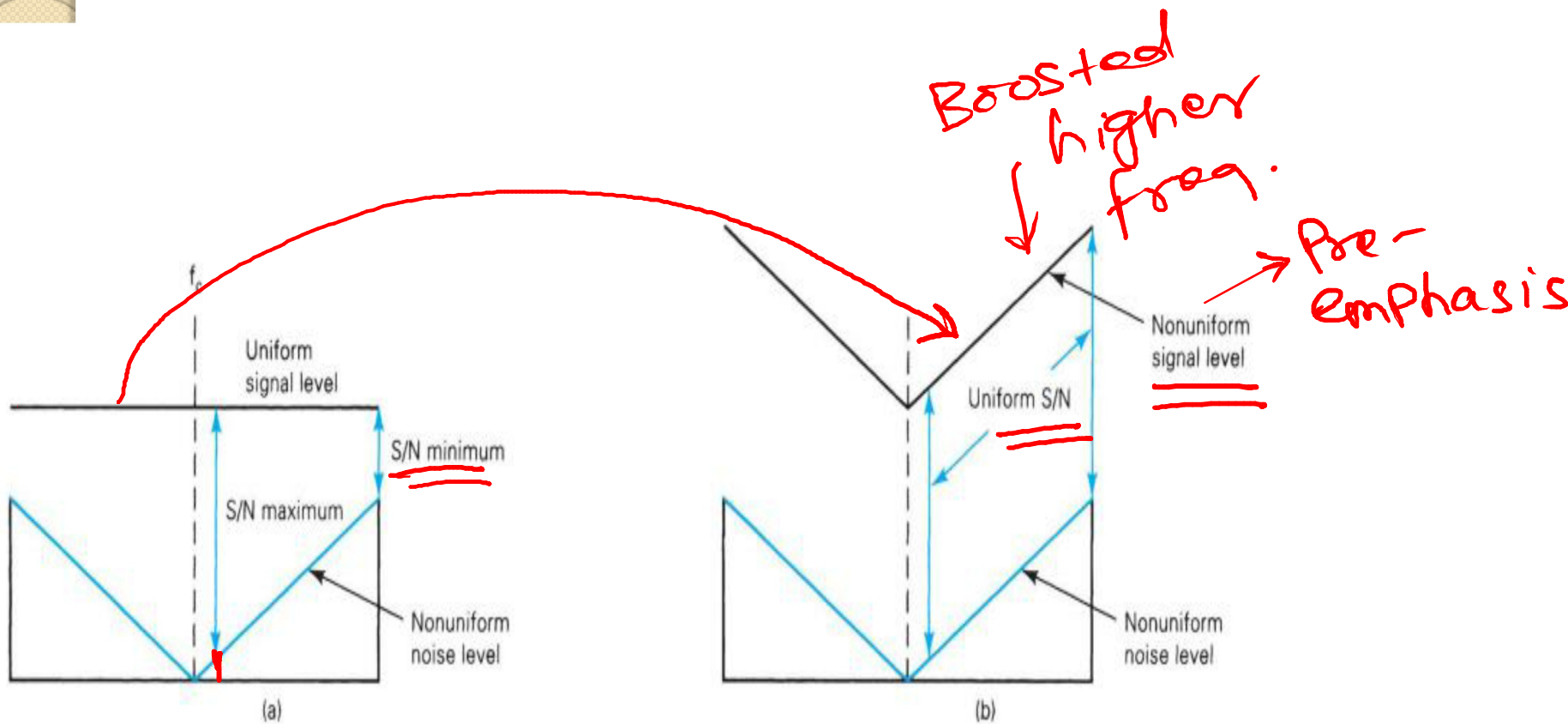
- When this unwanted carrier deviation is demodulated,
- It becomes noise if it has frequency components that fall in original signal
- The noise voltage of PM demodulator remains **constant** with frequency
- But for **FM** demodulator it increases **linearly** with frequency

FM Noise Triangle

- A plot of S/N vs. frequency



Pre-emphasis and De-emphasis Network



Pre-emphasis and De-emphasis Network

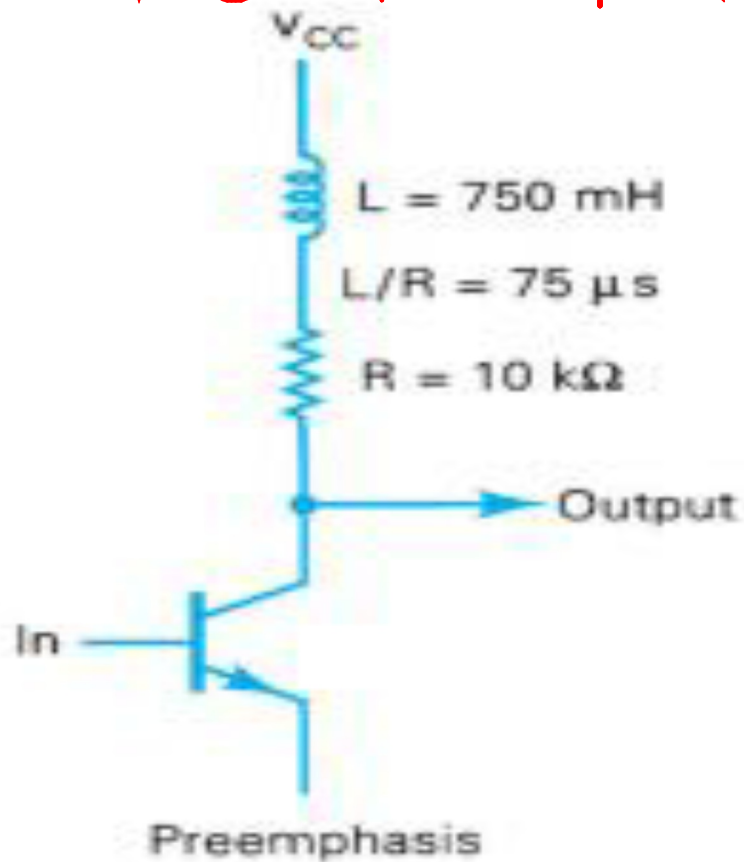
- The effect of noise increases with increase in frequency → SNR becomes poor at high frequency
- High frequency components of the message are badly affected by the noise
- **Pre-emphasis:** The effect of noise can be reduced by increasing value of modulation index

Pre-emphasis and De-emphasis Network

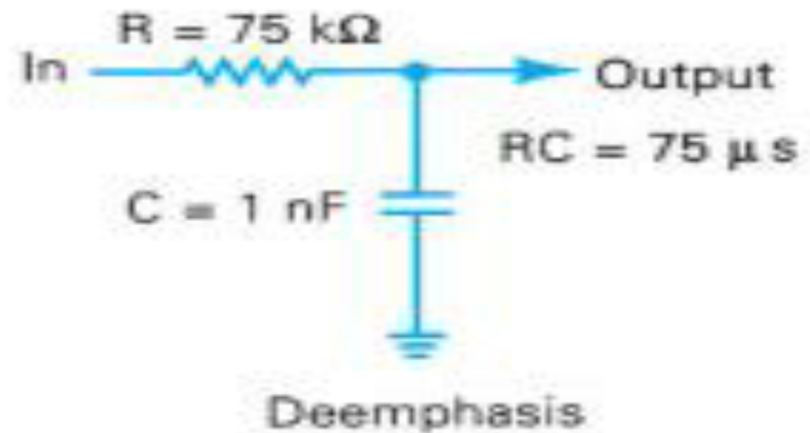
- This can be done by increasing the deviation (Δf) $\rightarrow$ increasing amplitude of modulating signal at high frequency
- The artificial boosting of higher modulating frequencies
- The boosted signal are brought to their original amplitude $\rightarrow$ **de-emphasis circuit**

Pre-emphasis and De-emphasis Network

High Pass Filter



Low Pass Filter



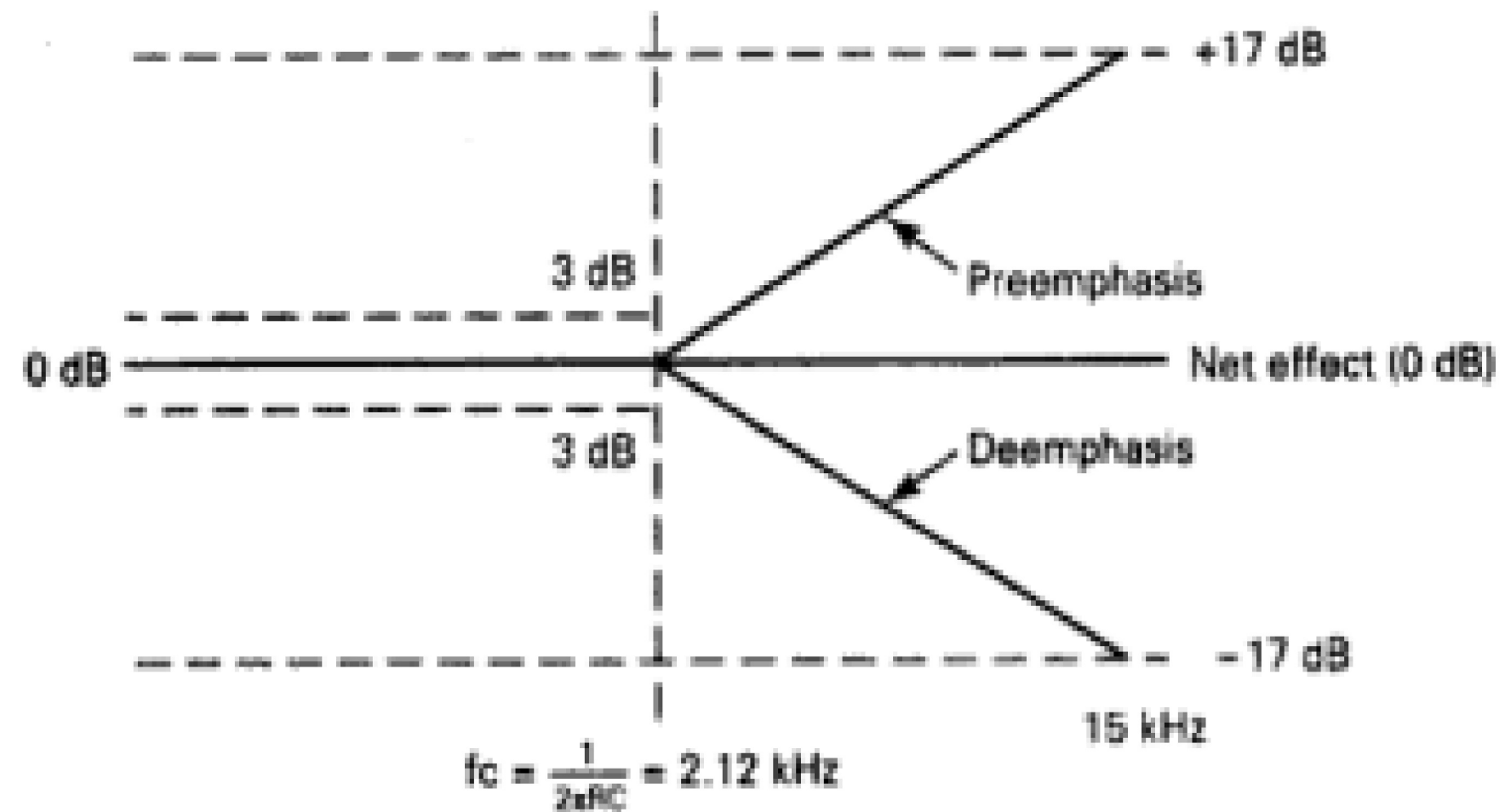
(a)

Pre-emphasis and De-emphasis Network

$$f_c = \frac{1}{2\pi RC}$$

$$f_c = \frac{1}{2\pi L/R}$$

Pre-emphasis and De-emphasis Network



(b)